AIRBNB MARKET INTELLIGENCE ANALYSIS

Data engineering internship assessment

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1. EXECUTIVE SUMMARY

This project presents a data engineering and exploratory analysis of the New York City Airbnb dataset obtained from the Inside Airbnb platform. The objective was to transform raw Airbnb data into meaningful business insights through data cleaning, profiling, and analysis.

The project involved loading multiple Airbnb datasets, assessing data quality, identifying missing values, cleaning pricing information, and performing exploratory data analysis on room types, neighbourhoods, review ratings, host characteristics, and listing density.

The analysis identified several important trends. Hotel rooms were found to have the highest average nightly price, while private and shared rooms represented lower-cost accommodation options. Premium neighbourhoods such as Fort Wadsworth, Civic Center, and Tribeca exhibited significantly higher prices compared to other locations. Listings with higher review ratings generally commanded higher prices, suggesting that customer satisfaction contributes to pricing power.

The findings highlight the importance of location, accommodation type, and service quality in determining Airbnb pricing. Based on these insights, several recommendations were developed for hosts, investors, and platform stakeholders.

2. OBJECTIVES AND SCOPE

The primary objective of this project was to demonstrate data engineering and analytical skills using a real-world Airbnb dataset.

The specific objectives were:

- To acquire and profile Airbnb listing data.
- To assess data quality and identify missing values.
- To clean and transform raw data into an analysis-ready format.
- To perform exploratory data analysis on pricing, location, room type, and host characteristics.
- To generate actionable business insights and recommendations.

This analysis was limited to the New York City Airbnb dataset. The project focused on data engineering, data quality assessment, and exploratory analysis rather than advanced machine learning techniques.

3. METHODOLOGY

The project followed a structured data engineering and analytics workflow to transform raw Airbnb data into meaningful business insights.

3.1 Data Acquisition

The dataset was obtained from the Inside Airbnb platform and included listing information, calendar data, review data, and neighbourhood information for New York city.

3.2 Data Profiling

Initial profiling was performed to understand the structure of the dataset, identify data types, determine record counts, and assess overall data quality.

3.3 Data Cleaning

The dataset was examined for duplicate records, missing values, and inconsistencies. Pricing information was standardized and converted into a numeric format suitable for analysis.

3.4 Exploratory Data Analysis

Exploratory analysis was conducted to identify patterns and relationships within the data. Key areas of analysis included pricing, room types, neighbourhoods, review ratings, host characteristics, and listing density.

3.5 Business Interpretation

Analytical results were translated into business insights and recommendations relevant to Airbnb hosts, investors, and platform stakeholders.

4. DATA ENGINEERING APPROACH

The project applied several data engineering practices to prepare the dataset for analysis.

4.1 Data loading

The Airbnb datasets were loaded into Python using the Pandas library. Multiple CSV files were imported and inspected to understand their structure and contents.

4.2 Data profiling

Dataset profiling was performed to identify the number of records, available attributes, data types, and potential data quality issues.

Key Dataset Statistics:

- Total Rows: 35,036
- Total Columns: 90
- Memory Usage: 24.1 MB

4.3 Missing value analysis

Several columns contained substantial missing values. The columns beds, bathrooms, license, host_about, and price contained the highest number of missing records.

4.4 Duplicate analysis

The dataset was checked for duplicate records. No duplicate records were identified, indicating a high level of uniqueness within the listing data.

4.5 Price standardization

The price column originally contained currency symbols and formatting characters. These were removed, and the values were converted into numeric format to support statistical analysis and business reporting.

```python
import pandas as pd

# Load cleaned dataset
listings = pd.read_csv("processed_data/listings_cleaned.csv")

print("=== PRICE COLUMN BEFORE CLEANING ===")
print(listings["price"].head())

# Remove $ and commas
listings["price"] = (
    listings["price"]
    .astype(str)
    .str.replace("$", "", regex=False)
    .str.replace(",", "", regex=False)
)

# Convert to numeric
listings["price"] = pd.to_numeric(
    listings["price"],
    errors="coerce"
)

print("\n=== PRICE COLUMN AFTER CLEANING ===")
print(listings["price"].head())

print("\n=== PRICE STATISTICS ===")
print(listings["price"].describe())

# Save dataset
listings.to_csv(
    "processed_data/listings_price_cleaned.csv",
    index=False
)

print("\nPrice cleaning completed successfully!")
```

Figure 1: Python code used to clean and standardize Airbnb listing prices.

4.6 Clean dataset creation

A cleaned version of the dataset was created and stored for use in subsequent analysis stages.

5. EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis (EDA) was conducted to understand key patterns within the Airbnb marketplace and identify factors influencing listing prices and market behavior.

The analysis focused on:

- Room Type Analysis
- Neighbourhood Price Analysis
- Review Rating Analysis
- Superhost Analysis
- Listing Density Analysis

The following sections present the findings and business implications of each analysis.

5.1 Room Type Analysis

Objective :

To analyze how Airbnb prices vary across different room types

Results :

Room Type	**Average Price $()**
Hotel Room	814.35
Entire Home/Apt	297.97
Private Room	184.76
Shared Room	178.02

Findings

- Hotel rooms have the highest average nightly price.
- Entire homes/apartments are the second most expensive accommodation type.
- Private rooms and shared rooms offer lower-cost alternatives for travelers.

Business Interpretation

Accommodation type significantly influences pricing. Premium accommodation options such as hotel rooms command substantially higher prices compared to shared and private room offerings.

5.2 Neighbourhood Analysis

Objective :

To identify the most expensive neighbourhoods in New York City.

Results :

Rank	**Neighbourhood**	**Average Price ($)**
1	Fort Wadsworth	1010.33
2	Civic Center	862.82
3	Tribeca	841.77
4	Navy Yard	726.83
5	Riverdale	716.17
6	Country Club	647.07
7	Financial District	601.81
8	Battery Park City	599.63
9	Brooklyn Heights	563.25
1o	Flatiron District	550.64

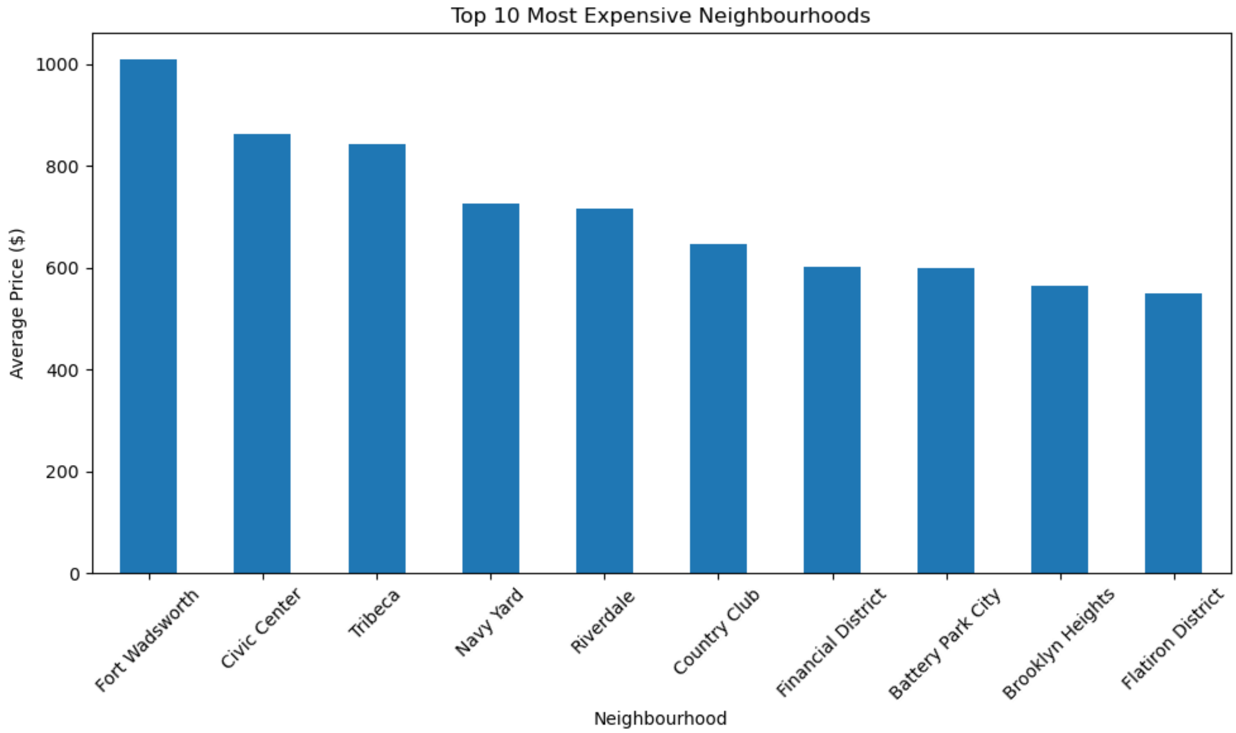

Figure 2: Top 10 Most Expensive Neighbourhoods in New York City

Findings

- Fort Wadsworth recorded the highest average Airbnb price.
- Several premium Manhattan neighbourhoods appeared among the most expensive locations.
- Location has a strong influence on listing prices.

Business Interpretation

Neighbourhood selection plays a major role in revenue generation. Properties located in premium areas can command significantly higher nightly rates.

03_price_cleaning.py | 05_neighbourhood_analysis.py | 06_neighbourhood_chart.py

src > 05_neighbourhood_analysis.py > ...

```python
import pandas as pd

listings = pd.read_csv("processed_data/listings_price_cleaned.csv")
listings = listings.dropna(subset=["price"])

neighbourhood_prices = (
    listings.groupby("neighbourhood_cleansed")["price"]
    .mean()
    .sort_values(ascending=False)
    .head(10)
)

print("=== TOP 10 MOST EXPENSIVE NEIGHBOURHOODS ===")
print(neighbourhood_prices)
```

Figure 3: Python code used to identify the top 10 most expensive neighbourhoods.

5.3 Review Rating vs Price Analysis

Objective :

To examine whether higher-rated listings are associated with higher prices.

Results :

Rating Group	**Average Price ($)**
0-3.0	183.07
3.0-4.0	198.78
4.0-4.5	215.76
4.5-5.0	234.24

Findings

- Average price increases consistently as review ratings improve.
- Listings with ratings between 4.5 and 5.0 achieve the highest average prices.
- Poorly rated listings tend to charge lower prices.

Business Interpretation

Customer satisfaction appears to contribute to pricing power. Hosts with excellent guest reviews may be able to charge higher nightly rates.

5.4 Superhost Analysis

Objective :

To investigate whether Superhost status influences pricing.

Results :

Host Type	**Average Price ($)**
Non-Superhost	254.11
Superhost	248.48

Findings

- Non-superhosts have a slightly higher average price than superhosts.
- The difference between the two groups is relatively small.

Business Interpretation

Superhost status alone does not appear to increase listing prices. Other factors such as location, property type, and guest demand may have a greater influence on pricing.

5.5 Listing Density Analysis

Objective :

To identify neighbourhoods with the highest concentration of Airbnb listings.

Results :

Rank	Neighbourhood	Number of Listings
1	Bedford-Stuyvesant	2408
2	Midtown	2037
3	Williamsburg	1958
4	Harlem	1637
5	Hell's Kitchen	1508
6	Upper West Side	1450
7	Upper East Side	1355
8	Bushwick	1278
9	Crown Heights	1054
10	East Village	893

Findings

- Bedford-Stuyvesant contains the largest number of Airbnb listings.
- Midtown and Williamsburg also represent highly concentrated Airbnb markets.
- Listing distribution varies considerably across neighbourhoods.

Business Interpretation

Neighbourhoods with large numbers of listings are highly competitive markets. Hosts may need stronger differentiation strategies to attract guests.

6. BUSINESS FINDINGS

Key business findings

Finding 1: accommodation type influences pricing

Hotel rooms recorded the highest average nightly price, while private and shared rooms were significantly cheaper. Accommodation type is therefore a major driver of Airbnb pricing.

Finding 2: location strongly affects revenue potential

Premium neighbourhoods such as Fort Wadsworth, Civic Center , and Tribeca command substantially higher prices than other areas, demonstrating the importance of location.

Finding 3: higher ratings support higher prices

Listings with higher review ratings consistently achieved higher average prices, suggesting that guest satisfaction contributes to pricing power.

Finding 4: Superhost status has limited impact on pricing

The analysis found only a small difference between Superhosts and Non-Superhosts, indicating that Superhost status alone may not justify premium pricing.

Finding 5: Market competition varies across Neighbourhoods

Neighbourhoods such as Bedford-Stuyvesant and midtown contain large numbers of listings, creating highly competitive market conditions.

7. BUSINESS RECOMMENDATIONS

Recommendation 1: focus on guest satisfaction

Hosts should prioritize cleanliness, communication, and overall guest experience to improve review ratings and support higher pricing.

Recommendation 2: optimize location-based pricing

Hosts operating in premium neighbourhoods should adopt pricing strategies that reflect local market demand and property value.

Recommendation 3: differentiate in competitive markets

Hosts in highly saturated neighbourhoods should focus on unique amenities, quality service, and attractive listing descriptions to stand out from competitors.

Recommendation 4: consider property type when setting prices

Pricing strategies should account for accommodation type, as hotel rooms and entire homes generate significantly higher average prices than shared accommodations.

Recommendation 5: use data-driven decision making

Airbnb hosts and property investors should continuously monitor market trends, pricing patterns, and customer feedback to improve profitability and competitiveness.

8. LIMITATIONS

This project has several limitations that should be considered when interpreting the results.

- The analysis was limited to the New York City Airbnb dataset and may not represent trends in other cities or regions.
- Several attributes contained substantial missing values, which reduced the amount of usable information for certain analyses.
- The analysis focused primarily on descriptive and exploratory techniques rather than predictive modeling.
- External factors such as seasonality, local events, economic conditions, and tourism trends were not considered.
- The dataset represents a specific point in time and may not fully capture changes in the Airbnb marketplace over time.

9. FUTURE IMPROVEMENTS

Several enhancements could be implemented in future versions of this project.

- Conduct multi-city comparisons to identify regional Airbnb trends.
- Develop predictive models to estimate listing prices and revenue.
- Build interactive dashboards using Power BI or Tableau for business users.
- Automate data ingestion and transformation using data pipelines.
- Incorporate additional external datasets such as tourism statistics and economic indicators.
- Deploy the solution in a cloud environment to support scalable analytics.

10. REFLECTION

This project provided valuable experience in applying data engineering and analytical techniques to a real-world dataset.

Throughout the project, I gained practical experience in data loading, profiling, cleaning, transformation, and exploratory data analysis using Python and Pandas. Working with a large dataset containing missing values and inconsistent formats helped strengthen my problem-solving and data preparation skills.

The project also improved my ability to translate technical analysis into meaningful business insights and recommendations. Understanding how factors such as location, room type, and customer ratings influence Airbnb pricing reinforced the importance of data-driven decision-making.

Overall, this project enhanced both my technical and analytical capabilities and provided hands-on experience relevant to a Data Engineering role.

The project successfully demonstrated fundamental data engineering practices, including data ingestion, cleaning, transformation, analysis, and business-oriented reporting using real-world Airbnb data.

11. AI USAGE DISCLOSURE

AI Tools Used

- ChatGPT

Purpose of use

- Obtain guidance on python implementation and debugging.
- Support interpretation of analytical results.
- Improve the clarity and presentation of written content.

Validation process

- All generated code was reviewed and executed by the author.
- Analytical results were validated against actual dataset outputs.
- Findings and recommendations were based on observed results from the dataset.

Human oversight

Final decisions regarding data cleaning, analysis, interpretation, and reporting were made by the author. AI-generated suggestions were reviewed, modified where necessary, and adapted to project requirements.

12. REFERENCES

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